

Chapter 10: Carboxylic Acids and Its Derivatives

PSPM 2011 / 2012

- Q. 1 (a) Arrange the following compounds in descending order of acidity. 5 m
Explain your answer.
phenol, cyclohexanol, benzoic acid, 4-nitrophenol

Answer (a) Benzoic acid > 4-nitrophenol > phenol > cyclohexanol

- $\text{C}_6\text{H}_5\text{COO}^-$ is the most stable since the negative charge is dispersed or delocalised on two electronegative oxygen atom / resonance.
Or
- Carboxy group is strong EWG. O-H bond became weaker. Then, H^+ is easily released
- $\text{NO}_2-\text{C}_6\text{H}_5\text{O}^-$ is less stable than $\text{C}_6\text{H}_5\text{COO}^-$ since the negative charge is dispersed on one oxygen atom and less electronegative carbon atom, and dispersed farther by electron withdrawing group of NO_2 .
Or
- $\text{C}_6\text{H}_5\text{O}^-$ is less stable than $\text{NO}_2-\text{C}_6\text{H}_5\text{O}^-$ since it has no electron withdrawing group - NO_2 .
 $\text{C}_6\text{H}_{11}\text{O}^-$ is the least stable since the negative charge is localized on one oxygen atom only.

PSPM 2012 / 2013

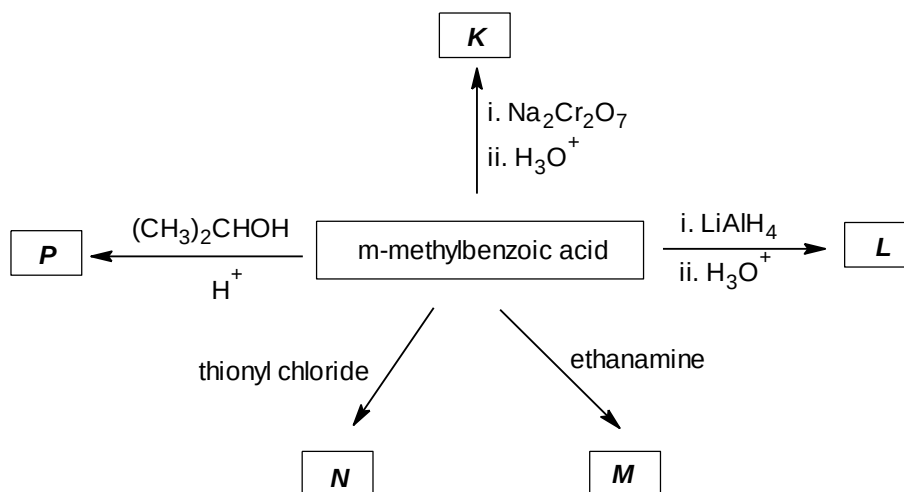
- Q. 2 (a) Arrange the following compounds in ascending order of acidity. Explain your answer. 5 m
2,3-dichlorobutanoic acid, 1-butanol, butanoic acid,
4-chlorobutanoic acid, 2-chlorobutanoic acid

Answer (a) 1-butanol < butanoic acid < 4-chlorobutanoic acid < 2-chlorobutanoic acid < 2,3-dichlorobutanoic acid
Note: Arrangement : Butanol vs acid - 1M
Between acids - 1M
Explanation:
Butanol vs acid : the strength of **O-H bond in acid is weakened** due to **inductive effect @ the carboxylate ion is stabilized by delocalisation / resonance @ O-H bond is strengthened due to inductive effect (EDG)**.
Position of halogen : the closer the halogen to carboxyl group the stronger the inductive effect, stronger the acid.
Number of halogen : More EWG, stronger the inductive effect, stronger the acid.

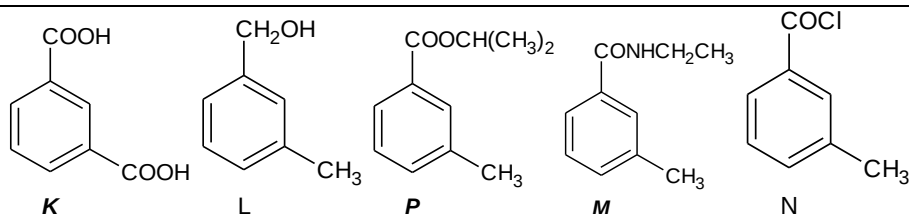
PSPM 2013 / 2014

- Q. 3 (a) Based on the reaction scheme in **FIGURE 1**, draw the structures of **K**, **L**, **M**, **N** and **P**.

5 m

**FIGURE 1**

Answer



PSPM 2014 / 2015

- Q. 4 (a) Methanoic acid is the only carboxylic acid that reacts with Tollens' reagent. Give your reasons. State the observation and write the chemical equation involved.

4 m

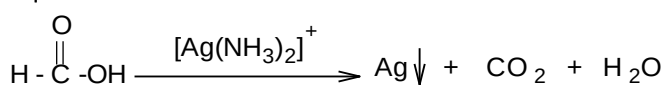
Answer

(a)

Methanoic acid has carbonyl functional/aldehydic/aldehyde groups thus it shows reducing properties in reaction with Tollens' reagent.

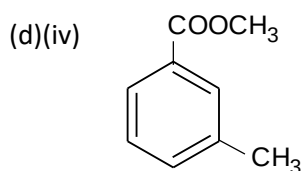
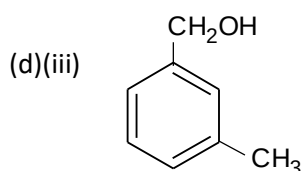
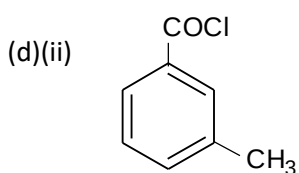
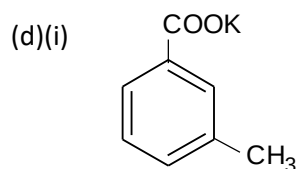
Observation: Silver mirror formed @ shiny grey precipitate @ silver or grey precipitate

Equation:



- Q. 4 (b) Predict the product of the reaction of 3-methylbenzoic acid with each of the following reagents:
- (i) KOH
 - (ii) SOCl_2
 - (iii) LiAlH_4 followed by H_3O^+
 - (iv) CH_3OH , H^+ / Δ

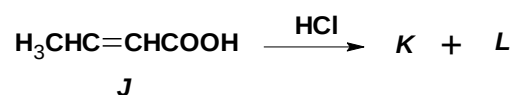
Answer



PSPM 2015 / 2016

- Q. 5 (a) Reaction of an unsaturated carboxylic acid with hydrochloric acid yields compounds K and L as shown below:

8 m



- (i) Draw the structures of K and L.
- (ii) Which product is more acidic and give your reason.
- (iii) Predict the product forms when J reacts with potassium metal.

Answer (a)(i) **K**: $\text{CH}_3\text{CH}_2\overset{\text{Cl}}{\text{CH}}\text{COOH}$

L: $\text{CH}_3\overset{\text{Cl}}{\text{CH}}\text{CH}_2\text{COOH}$

(a)(ii) $\text{CH}_3\text{CH}_2\overset{\text{Cl}}{\text{CH}}\text{COOH}$ is more acidic than $\text{CH}_3\overset{\text{Cl}}{\text{CH}}\text{CH}_2\text{COOH}$

Reason :

The **presence of Cl increase the acidity due to inductive effect.**

@

The presence of EWG stabilised carboxylate ion

The **acidity decrease** as the **distane between Cl and carboxyl group increase.**

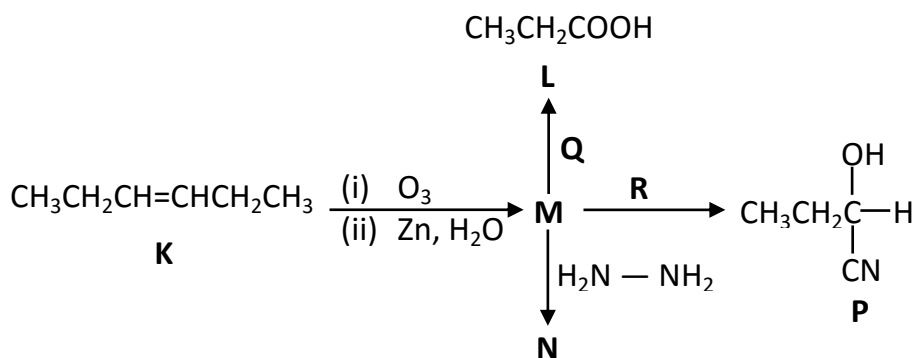
(a)(iii) $\text{CH}_3\text{CH}=\text{CHCOOK}$

PSPM 2016 / 2017

Q. 6

Consider the following reaction scheme.

15 m



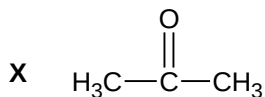
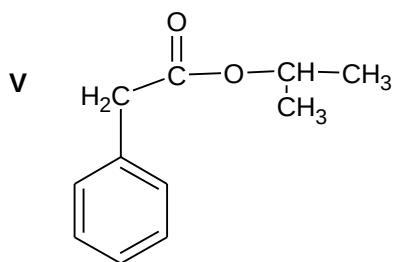
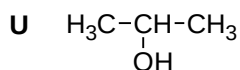
- Draw the structures of **M** and **N** and give the reagents **Q** and **R**.
- Suggest the products formed when **L** reacts with:
 - SOCl_2
 - LiAlH_4 followed by hydrolysis
 - sodium metal
 - NH_3 / Δ
- Arrange **K**, **L** and **M** in order of increasing boiling point. Explain.
- State the observation and write the chemical equation when **M** reacts with Tollen's reagent. Suggest **one(1)** chemical test that can be conducted to confirm the presence of functional group in **M**.

Answer	(a) $M = \text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{H}$ $N = \text{CH}_3\text{CH}_2\overset{\text{NNH}_2}{\underset{\parallel}{\text{C}}}\text{H}$ Q = $\text{KMnO}_4, \text{H}^+, \Delta$ @ $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+, \Delta$ @ i) $\text{KMnO}_4, \text{OH}^-, \Delta$ ii) H_3O^+ R = KCN, H^+ @ NaCN, H^+
(b)(i)	$\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{Cl}$
(b)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
(b)(iii)	$\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{O}^-\text{Na}^+ \text{ @ } \text{CH}_3\text{CH}_2\text{COONa}$
(b)(iv)	$\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{NH}_2$
(c)	M < K < L L – carboxylic acid has hydrogen bonding M – aldehyde has dipole-dipole interaction K – alkene has van der Waals @ London forces K has bigger size @ larger molecular weight @ larger contact surface area than M
(d)	Observation : Silver mirror forms Chemical equation: $\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{H} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 3\text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\parallel}{\text{C}}}\text{O}^- + \text{Ag}$ Chemical test : Brady's test

PSPM 2018 / 2019

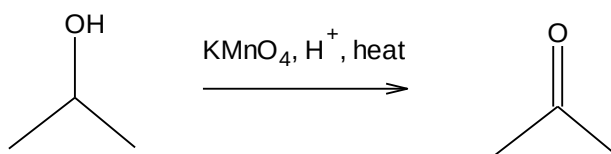
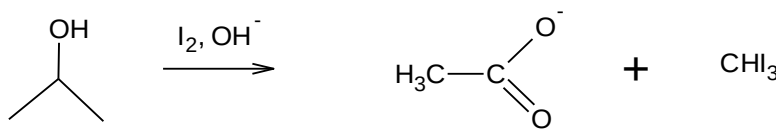
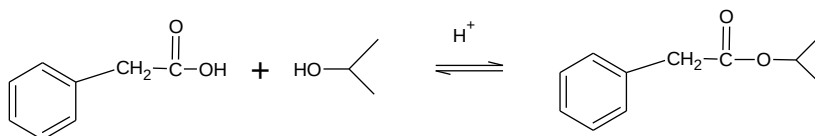
- Q. 7 (a) Treatment of 2-phenylethanoic acid with alcohol U gives an ester V with molecular formula of $C_{11}H_{14}O_2$. Alcohol U reacts with alkaline iodine solution to form a yellow precipitate W, whereas oxidation of U yields X. 10m
- (i) Draw the structures for U, V and X.

Answer



- (a)(ii) Write all the chemical equations involved.

Answer



- (a)(iii) Gives the IUPAC name of W.

Answer CHI_3 – triiodomethane @ 1,1,1-triiodomethane

Q. 9 (b) Explain why methanoic acid gives a positive test towards Tollens' reagent but ethanoic acid does not. Write the equation for this reaction.

Answer

Methanoic acid contains **methanoyl group @ -CHO group @ aldehyde group**. Hence, it has **reducing property @ can be oxidized**.

